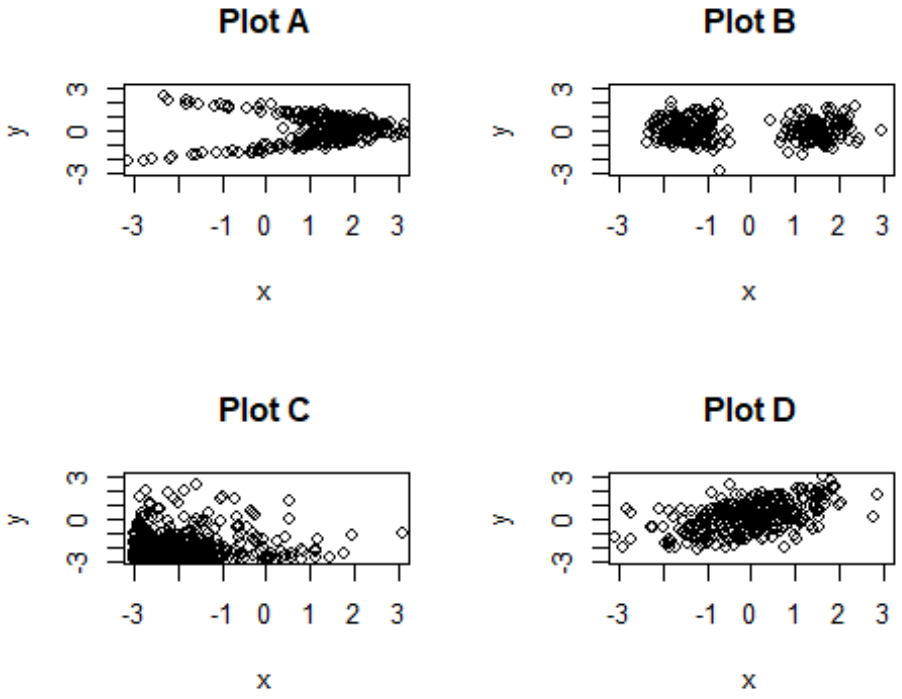
Review Questions for Lecture 5 and 6

1. In which of the following plots are x and y associated?

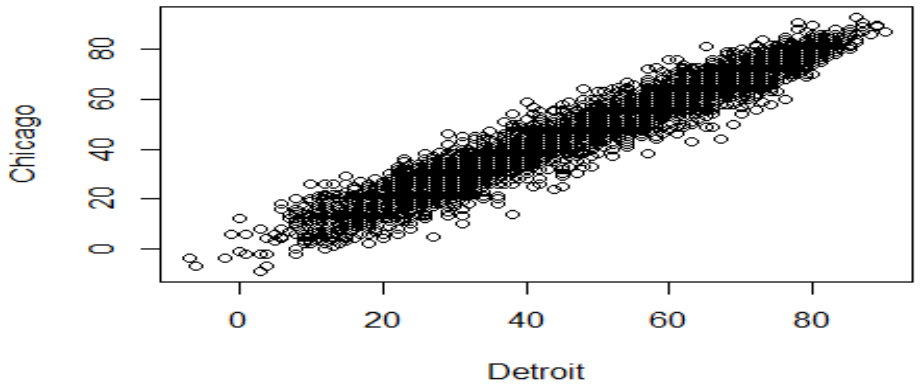


2. Using formulas shown in class, show that

$$\text{corr}(x, y) = \frac{\overline{xy} - \bar{x}\bar{y}}{\sqrt{\left(\overline{x^2} - \bar{x}^2\right)\left(\overline{y^2} - \bar{y}^2\right)}}$$

Questions 3-7 refer to the scatterplot below. The scatterplot shows average daily temperatures in Detroit and Chicago over 10 years. Temperatures are measured in Fahrenheit and the correlation is 0.97. The conversion between Celsius and Fahrenheit is as follows:

$$F = \frac{9}{5}C + 32$$



3. If you converted the temperatures to Celsius, what would be the new correlation?

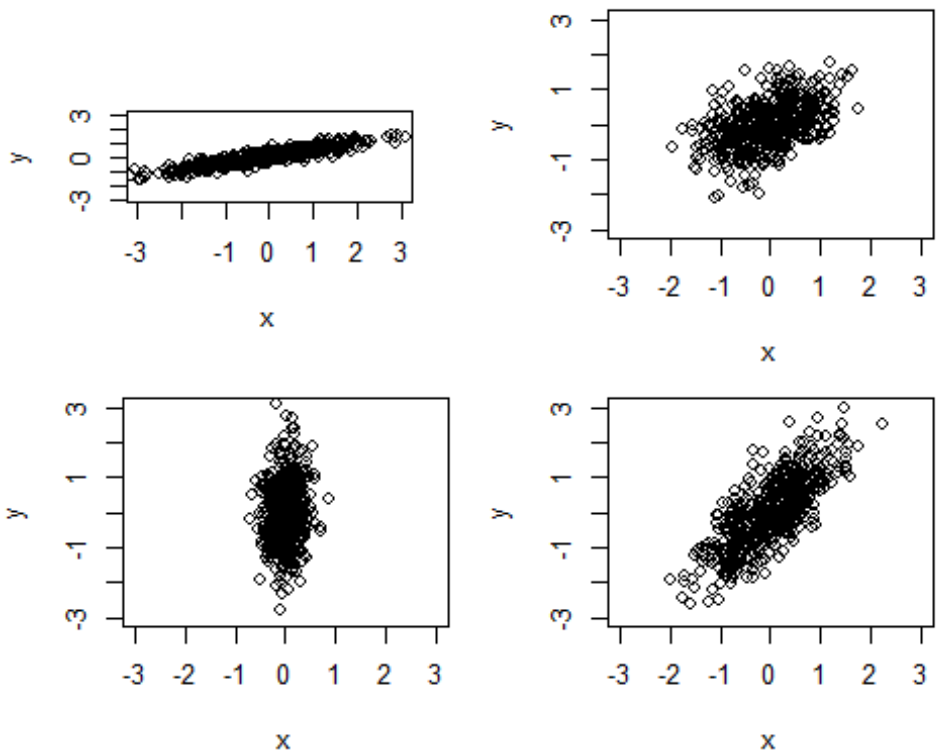
4. If someone accidentally converted Detroit's temperatures to Celsius, but left Chicago's in Fahrenheit, what would be the new correlation?

5. From the graph, visually estimate the SD of x (the Detroit data) and the SD of y (the Chicago data). One rough estimate of the standard deviation is *range/4*. This is because most values will be within 2 standard deviations of the mean if the temperature histograms roughly follows the normal curve.

6. Approximately what is the covariance between x and y? (Hint: Use your answer from the previous question.)

7. If someone converted both temperatures to Celsius, what would be the new covariance?

8. There are four scatterplots below. One has correlation 0, one has correlation 0.5, one has correlation 0.75, and one has correlation 0.9. Which is which?



9. Correlation is a good measure of the strength of an association when the data is football shaped. Give three examples of data where correlation is not a good measure of association.

10. In a large representative population, the correlation between height and weight of people age 8 is a. The correlation between height and weight of people age 12 is b. The correlation between height and weight for everyone ages 8 through 12 is c. Which of the following are true?

$$c > b$$

$$c < b$$

$$c \approx b$$

$$c > (a + b)/2$$

$$c < (a + b)/2$$

$$c \approx (a + b)/2$$